SENTINEL-X

Wi-Fi Based Indoor Assets Positioning and Tracking System

TAG 03 & TAG 04

Configuration, Access Point Setup and Detection Verification

Project Documentation

ESP32 Wi-Fi tag configuration and anchor-side detection results

1. Overview

Sentinel-X is a Wi-Fi-based indoor asset positioning and tracking system. In this implementation stage, ESP32 boards are configured to represent Wi-Fi tags. Each tag broadcasts its own wireless network identity. An anchor device scans for these networks and reads the received signal strength indicator (RSSI), which provides a basic indication of the tag's signal level at the anchor.

2. Working Principle

Wi-Fi tag: An ESP32 configured with a unique tag ID and SSID. It starts a Wi-Fi access point so that the tag can be identified during scanning.

Anchor: A receiving ESP32 that scans nearby Wi-Fi networks, identifies configured tag SSIDs, and records signal information.

RSSI: Received Signal Strength Indicator, measured in dBm. Values closer to zero normally indicate a stronger received signal. RSSI can fluctuate due to walls, distance, interference, and device orientation; therefore, it is a signal-strength measurement, not an exact distance by itself.

Detection workflow: The tag starts its access point → the anchor scans nearby Wi-Fi networks → the anchor matches the tag identity → the anchor reports RSSI and other available details on the Serial Monitor.

3. Hardware and Software Setup

The supplied screenshots show the Arduino IDE 2.3.10 with an ESP32 Dev Module selected and the Serial Monitor connected through COM6. The tag configuration output confirms that each board starts its access point and prints its SSID, IP address, MAC address, and Wi-Fi channel.

Parameter	TAG-03	TAG-04
Tag ID	TAG-03	TAG-04
Wi-Fi SSID	SX_TAG_03	SX_TAG_04
IP address	192.168.4.1	192.168.4.1
Wi-Fi channel	6	6
Tag MAC address	8C:94:DF:6D:99:5D	8C:94:DF:6C:9A:95
Startup status	Ready for detection	Ready for detection

Note: Both tags use the default ESP32 SoftAP address shown in the Serial Monitor. The MAC addresses distinguish the two devices.

4. TAG-04 Configuration and Startup

The following Arduino IDE capture shows the TAG-04 startup confirmation. The Serial Monitor reports that the access point started successfully and that TAG-04 is ready for detection.

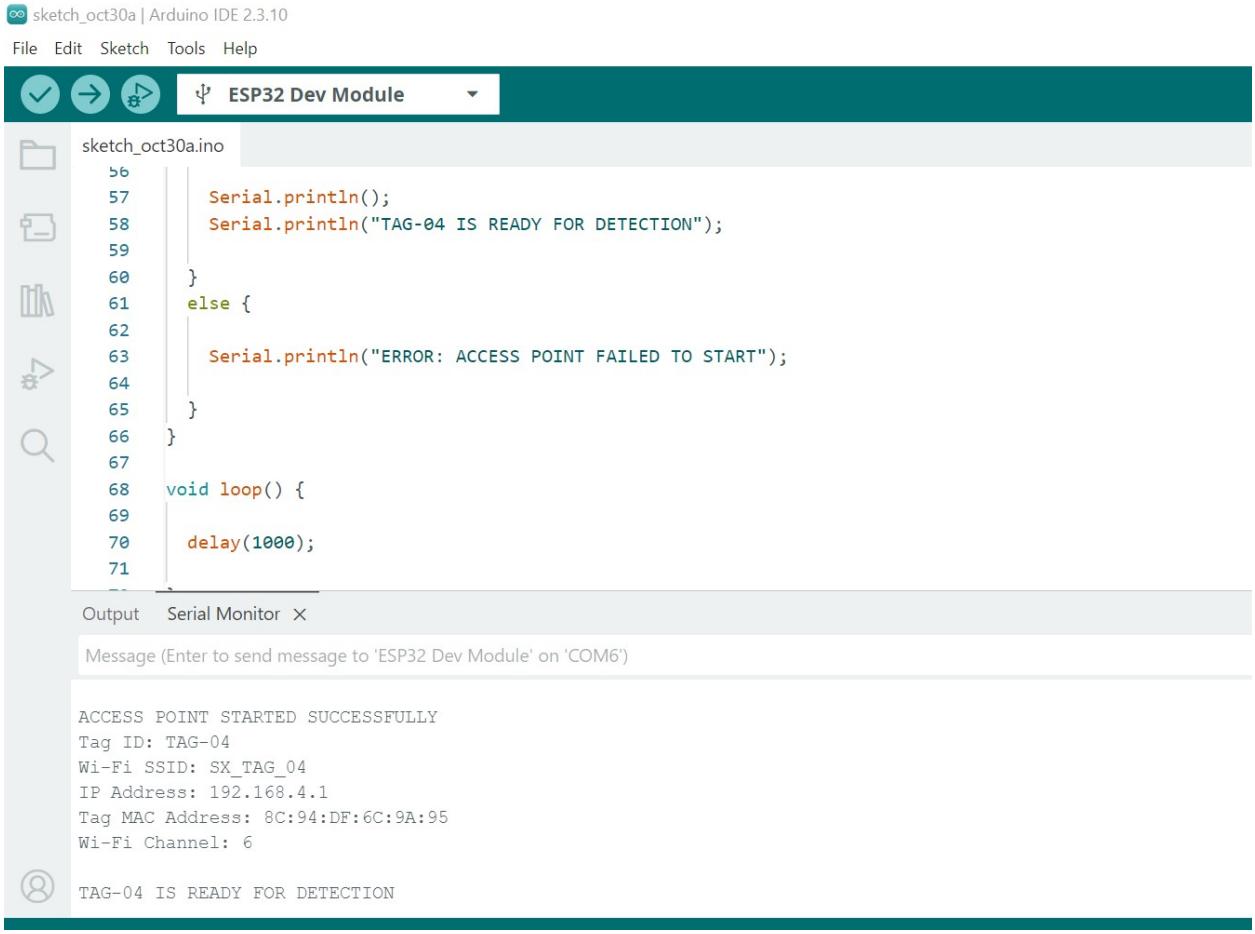


Figure 1. TAG-04 access point startup details in Arduino IDE Serial Monitor.

5. TAG-03 Configuration and Startup

The TAG-03 capture follows the same setup pattern with its own tag identity and SSID. The output confirms that TAG-03 is ready for detection.

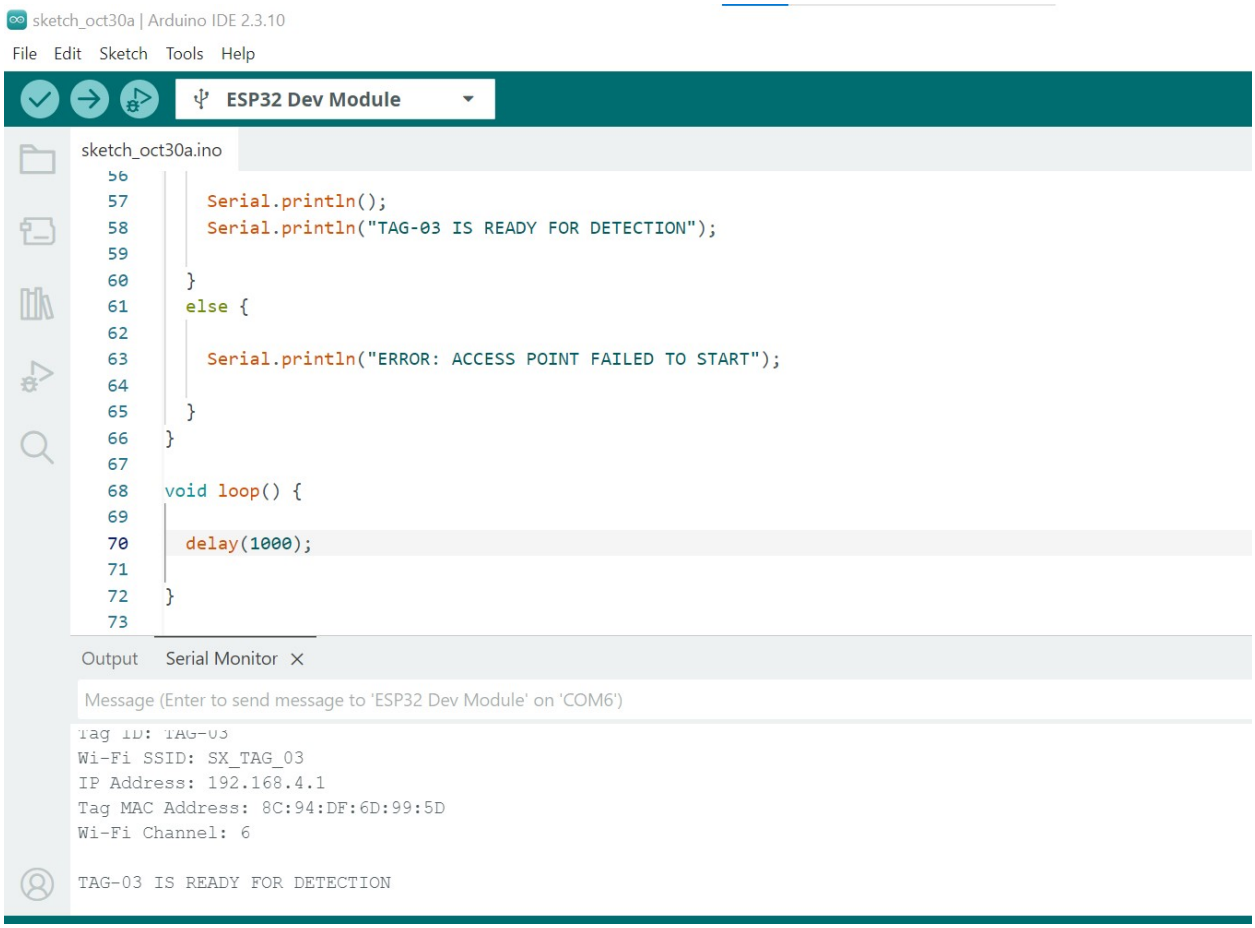


Figure 2. TAG-03 access point startup details in Arduino IDE Serial Monitor.

6. Anchor Detection and Verification

After both tag access points were started, the anchor (ANC-01) scanned the Wi-Fi environment. The supplied Serial Monitor capture shows successful detection records for TAG-04 and TAG-03. For each detected tag, the anchor printed the RSSI, Wi-Fi channel, and tag MAC/BSSID, followed by the message “Measurement completed.”

Detection field	TAG-04	TAG-03
Anchor ID	ANC-01	ANC-01
Detection status	Detected successfully	Detected successfully
RSSI	-54 dBm	-58 dBm
Wi-Fi channel	6	6
Tag MAC/BSSID	8C:94:DF:6C:9A:95	8C:94:DF:6D:99:5D
Measurement	Completed	Completed

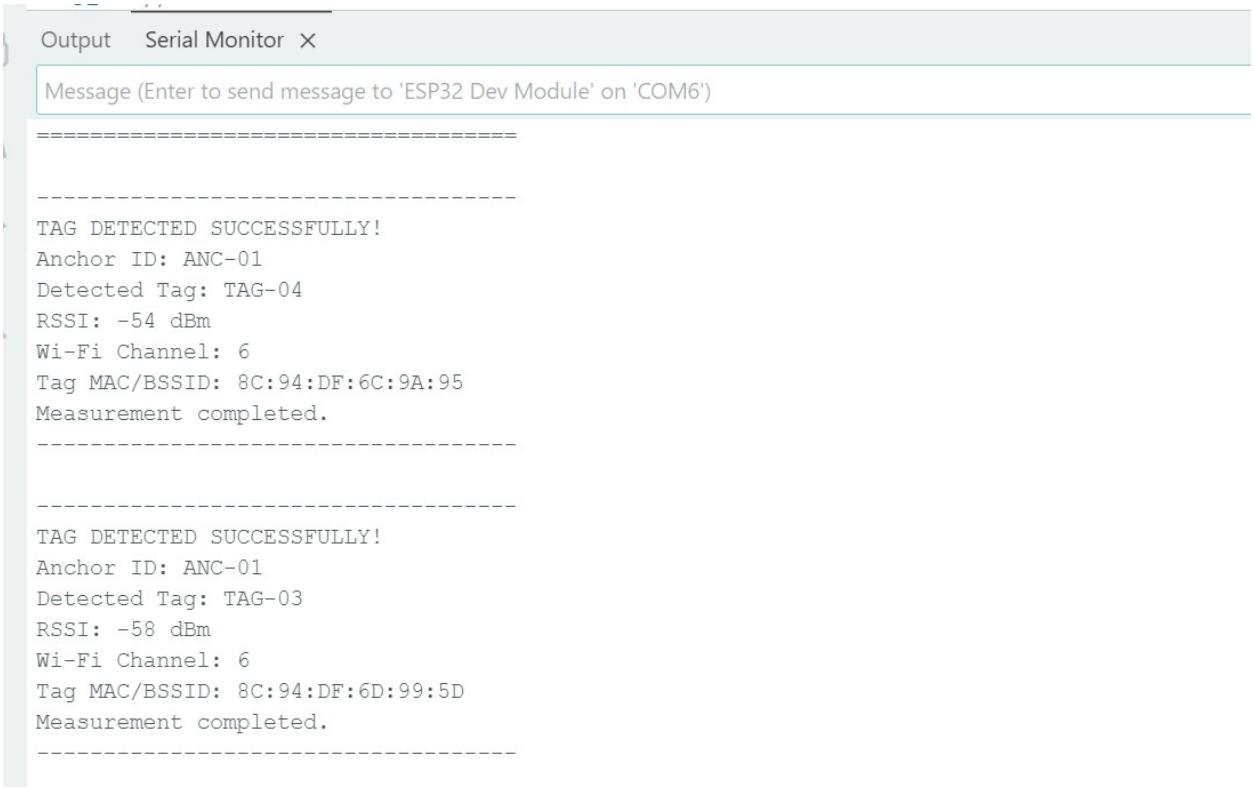


Figure 3. Anchor ANC-01 successfully detects TAG-04 and TAG-03 and records their RSSI values.

7. Results and Interpretation

The test confirms that both configured Wi-Fi tags were visible to the anchor and could be identified using their tag MAC/BSSID information. TAG-04 was recorded at -54 dBm, while TAG-03 was recorded at -58 dBm. In this particular measurement, TAG-04 had a signal level 4 dB stronger than TAG-03. This comparison applies only to the captured test; repeated measurements in different positions are needed for reliable position estimation.

8. Conclusion

TAG-03 and TAG-04 were configured as ESP32 Wi-Fi access points and successfully started on Wi-Fi channel 6. Anchor ANC-01 detected both tags and reported their identities and RSSI values. This validates the basic tag broadcasting and anchor detection stage of Sentinel-X and provides a foundation for later data collection, filtering, calibration, and indoor position estimation.